



(b) The floodlight (**X**) and audio alarm (**Y**) operate according to the following criteria:

Parameter	Description of parameter	Binary value	Condition
A	external daylight level	1	Low
		0	High
B	front door	1	Open
		0	Closed
C	person is within 2 m	1	Detected
		0	Not detected
D	back door	1	Open
		0	Closed
E	security system	1	Switched on
		0	Switched off

The floodlight turns on (**X = 1**) if:

- the security system is switched on
and
- the external daylight level is low
and
- a person is detected within 2 m.

The audio alarm turns on (**Y = 1**) if:

- the security system is switched on
and
- one **or** more doors are open, **or** a person is detected within 2 m.

Write logic expressions for the security system.

X =

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Y =

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[2]

(c) Explain whether the security system is an example of a monitoring system or a control system.

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[3]

